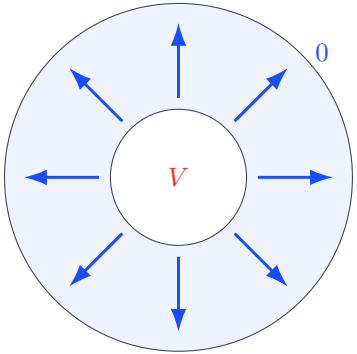
二维同轴电容的能量检验：场能与导体电荷给出同一单位长度电容



$$E_\rho = \frac{V}{\rho \ln(b/a)}$$

$$U = \int_a^b \frac{\varepsilon_0 E^2}{2}(2\pi\rho L)d\rho = \frac{\pi\varepsilon_0 L V^2}{\ln(b/a)}$$

$U = CV^2/2 \Rightarrow C = \frac{2\pi\varepsilon_0 L}{\ln(b/a)}$，与电荷积分一致。